Ruixiang Xue 薛瑞翔

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RESEARCH RESUME

Targeted for research and algorithm roles in point cloud compression, 3D Gaussian splatting compression, neural scene representation, driving world models, and AI multimedia systems.

EDUCATION

Nanjing University — 2021.09 - 2027.06

Ph.D. Student, Information and Communication Engineering

Hangzhou Dianzi University — 2017.09 - 2021.06

B.Eng., Electronic Information Engineering

WORK EXPERIENCE

Geely — 2026.05 - Present

Algorithm Intern - Working on driving world models, dynamic street-scene novel-view synthesis, and restoration of world-model or 3D scene generation artifacts.

- Studied and analyzed driving world model and street-view synthesis methods including Vista, StreetCrafter, SymDrive, FreeFix, and DriveFix.
- Built evaluation and deployment workflows around StreetCrafter and UniSplat-style baselines for dynamic street-scene generation experiments.
- Explored restoration training pairs and confidence or outlier filtering strategies for novel-view artifacts in driving scenes.

OPPO — 2024.02 - 2024.11

Algorithm Intern - Worked on intelligent point cloud compression research and MPEG AI-PCC standardization.

- Developed and evaluated intelligent point cloud compression algorithms around MPEG AI-PCC standardization.
- Implemented point cloud codec software and evaluated performance across standard test sequences.
- Participated in multiple MPEG meetings, submitted 8 standardization proposals, and filed 5 invention patents.

RESEARCH PROJECTS

Intelligent Point Cloud Compression

Research and development of deep learning based point cloud compression methods for MPEG AI-PCC.

- Designed Transformer models for point cloud representation and efficient spatial feature extraction.
- Used density changes during point cloud downsampling to guide reconstruction mode selection.
- Achieved 39% geometry compression gain and 34% attribute compression gain over the baseline under public test conditions.

3D Gaussian Splatting Coding

Compact and progressive compression for 3D Gaussian splatting scenes.

- Investigated compactification, compression, and reconstruction methods for 3D Gaussian splatting.
- Proposed RecastGS to reorganize pretrained 3DGS into a region-aware layered hierarchy with prompt-driven object extraction and progressive distillation.
- Proposed LayeredCGS, a feed-forward layered 3DGS compressor with cross-layer context modeling and truncatable bitstreams.
- Achieved 35% BD-Rate gain over the feed-forward FCGS baseline, with up to 2 dB higher ROI PSNR under comparable bitrates.
- The resulting paper, 3D Gaussian Splatting Compression with Object Scalability, has been accepted by ECCV 2026 and is not yet on arXiv.

Driving World Model and Street-Scene Restoration

Research exploration on dynamic street-scene generation, novel-view artifacts, and restoration pipelines for driving simulation.

- Surveyed Vista-style driving world models and StreetCrafter-style controllable street-view video diffusion methods.
- Compared restoration and refinement directions including Difix3D+, SymDrive, FreeFix, and DriveFix to identify gaps in artifact distribution, temporal consistency, and input requirements.
- Explored evaluation matrices for StreetCrafter and UniSplat baselines, including generated views, real references, and quantitative summaries.
- Focused on constructing degraded-clean pairs closer to inference-time novel-view artifacts and filtering unreliable pseudo supervision.

SELECTED PUBLICATIONS

3D Gaussian Splatting Compression with Object Scalability

Ruixiang Xue et al. (authors to be finalized after public release). *European Conference on Computer Vision (ECCV), 2026*. Accepted, not yet on arXiv

- Introduced RecastGS, a post-training method that reorganizes pretrained 3DGS into a region-aware layered hierarchy for object-level and ROI-adaptive quality control.
- Introduced LayeredCGS, a feed-forward layered 3DGS compressor that models cross-layer dependencies and generates truncatable bitstreams for fast preview and progressive refinement.

A Versatile Point Cloud Compressor Using Universal Multiscale Conditional Coding-Part I Geometry

Jianqiang Wang, Ruixiang Xue, Jiaxin Li, Dandan Ding, Yi Lin, Zhan Ma. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. SCI Q1, CCF-A, impact factor 18.6, co-first author

- Proposed a universal multiscale conditional point cloud geometry coding framework supporting lossy/lossless, static/dynamic, and diverse point cloud data.
- Outperformed MPEG G-PCC, V-PCC, and learning-based methods with second-level codec complexity; improved lossless compression by 30% and lossy compression by 92% overall.

NeRI: Implicit Neural Representation of LiDAR Point Cloud Using Range Image Sequence

Ruixiang Xue, Jiaxin Li, Tong Chen, Dandan Ding, Xun Cao, Zhan Ma. *IEEE International Conference on Acoustics, Speech, and Signal Processing 2024*. CCF-B, first author

- Proposed the first implicit neural representation method for LiDAR point cloud compression by converting 3D point clouds into 2D range images and coding them with spatiotemporal conditional neural networks.
- Reached 376 FPS real-time decoding and achieved 91% overall gain at low bitrates over MPEG G-PCC, HEVC, and other learning-based point cloud compression methods.

A Unified Point Cloud Compressor Using Universal Multiscale Conditional Coding-Part II Attribute

Ruixiang Xue et al. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. SCI Q1, CCF-A, impact factor 18.6, second author

GRNet: Geometry Restoration for G-PCC Compressed Point Clouds Using Auxiliary Density Signaling

Ruixiang Xue et al. *IEEE Transactions on Visualization and Computer Graphics*. SCI Q1, CCF-A, impact factor 6.5, second author

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SKILLS

Codec and Compression: Deep learning based point cloud coding, 3D Gaussian splatting coding, Driving world models, Street-view synthesis and restoration, MPEG AI-PCC standardization, G-PCC and V-PCC

Communication: CET-6, English presentations at international meetings, Technical literature search and reading

Programming: Python, PyTorch, Linux, AI coding workflows

AWARDS

First Prize Scholarship, Nanjing University, 2021 - 2025

Special Prize, 12th Zhejiang College Student Entrepreneurship Plan Competition, 2020